

Notes about some papers related to my Thesis

Miguel Ángel

August 2026

1 Paper: Learning to minimise regret in Route Choice

Main concepts introduced:

- **External regret of an agent:** It is used to evaluate the performance of a learning algorithm.

$$R_i^T = \max_{a_i^* \in A_i} \frac{1}{T} \sum_{t=1}^T r(a_i^*) - \frac{1}{T} \sum_{t=1}^T r(\hat{a}_i^t)$$

The first term considers the action that would have led to the highest average reward (computed after seeing all outcomes). The second term is the average reward obtained by the learning algorithm. The variable $\hat{a}_i^t$ denotes the action chosen by agent i at timestep t .

A problem with this metric, is that in our thesis context (same for the paper) it cannot be computed, since for the timestep t , the agent only knows the reward of the chosen action.

- **Estimated reward action:**

$$\hat{r}(a_i^t) = \begin{cases} r(a_i^t) & \text{if } a_i^t = \hat{a}_i^t, \\ \hat{r}(a_i^{t-1}) & \text{otherwise.} \end{cases}$$

Basically it says that if the action is the chosen action at timestep t , its estimate reward is the observed reward. If the action is non chosen at timestep t , then its estimate is the same than previous timestep estimate. Basically is using as estimate the last observed reward on that action.

- **History of estimates agent i :**

$$H_i = \{\hat{r}(a_i^t) \mid a_i^t \in A_i, t \in [1, T]\}$$

Basically this is a data structure that contains the reward estimates of all actions for all timesteps.

- **Estimated action regret of agent i for action a up to time T**

$$\hat{R}_{i,a}^T = \max_{b_i^t \in A_i} \frac{1}{T} \sum_{i=1}^T r(b_i^t) - \frac{1}{T} \sum_{i=1}^T r(a_i^t)$$

Basically it is measuring how much worse an action is compared to the best action (take the difference of the estimated avg reward of best action with the estimated avg reward of some action).

- **Estimated external regret of agent i**

$$\hat{R}_i^T = \max_{a_i^t \in A_i} \frac{1}{T} \sum_{i=1}^T \hat{r}(a_i^t) - \frac{1}{T} \sum_{i=1}^T \hat{r}(\hat{a}_i^t)$$

Basically is it comparing the estimate avg reward of the best action (benchmark) against the estimate avg reward obtained with the learning algorithm.

On this paper, timestep t , it means an episode. Because they are using volume delay functions, it is assuming that for each episode the reward of a route is the same for all agents. This is natural when using VDF models. When using a microsimulator, for each episode, the same route produces different rewards for each agent.

Then it builds a Q-learning, in which the updates of the action values uses this concepts and it proves it performs better than standard Q-learning.

2 Paper: Where does this road go?

Traditional route-choice models often assume that travelers already know all route options and their travel times. In reality people explore unfamiliar routes, learn from past experience, adjusts preferences based on feedback and new information. Authors propose using RL models to capture this process. They apply this models to driving simulator experiments and stated preferences surveys (people is asked which route they will choose in hypothetical situations).

- **Key findings:**

1. People explore routes: People do not always pick the currently best-known route. Sometimes try alternatives to gather information.
2. People learn from experience
3. Different people learn differently: Some adapt quickly, other learn more slowly
4. Information matters: Providing information significantly changes route choices (for example about route reliability...

- **Introduction ideas**

People do not know in advance which option they prefer. You need to gradually explore and learn which one you prefer. This may happen when:
1. Disruptions (road closure or traffic incident...). 2. New options appear (new bus route...). 3. New environment (moving to a new city...).

When someone is learning, their choices may look inconsistent, but stable preferences can emerge later. After enough experience uncertainty decreases, exploration becomes less necessary and choices become more stable. For example the week 1 you explore A, B and C. The week 2 you explore mostly A and B. The week 3 you explore almost always A.

- **Inertia** Inertia means sticking with what you already do. The authors asked how did that inertia arise. Inertia may be the result of learning not a fundamental behavioral trait. They argue that "the modelling and understanding of how travelers learn about their preferences through a process of exploring the options available to them is understudied."

Reasons why is understudied is lack of suitable data (to study learning you will need repeated observations of the same individual over time, and most travel datasets do not contain that information), lack of suitable models (many traditional models assume preferences are fixed, travelers already know all alternatives and choices are independent observations. A traveler in a DUE model is assumed to know all relevant alternatives and their costs (no traveler can improve their travel time by unilaterally switching routes)).

- **RL intuition** 1. Choose an action. 2. Observe outcome. 3. Compare outcomes with expectations. 4. Update future behavior. Prediction error = Observe reward - Expected reward
- **Criticism of RL** Artificially laboratory tasks that have limited real-world consequences (slot machine tasks...). A participant may lose or win a few dollars but nothing else. Travel decisions are different, they affect travel time, daily schedule...). If I know I have to go to work I might not explore because it has consequences. If I don't have to go to work, I may feel like exploring since nothing very bad will happen. I don't learn the same if environment has consequences than if it not.

- **Paper contributions**

1. **Trial-level RL model.** It means they use the entire sequence of choices taking into account the order. Traveler 1: AAAABBBB. Traveler 2: ABABABAB. A trial-level RL can distinguish between them.
2. **Estimate cognitive parameters:** Learning rate, choice stochasticity (how willing is someone to try uncertain alternatives), prior bias (initial preferences before any experience, prefer highways...)

3. **Combine experience and information:** Humans often trust more experience than descriptions. The learning is not the same when someone told you don't do this because you are gonna hurt yourself, than when you actually do it, and experience it yourself.

Example: Navigation app: Route B averages 15 min. You may ignore it. But after driving and seeing 15 min your preference changes.

- **Model proposed in the paper**

1. For each route i , the traveler has an internal estimate $Q_i(t)$. How good do I currently think route i is?
2. Update rule: $Q_{new} = Q_{old} + \alpha$ (observed - expected)
3. Learning rate α : It tells us how much weight is given to new experiences
4. How do travelers choose routes? It uses a softmax choice rule. The intuition is that route with lower travel times get higher probability.

$$P_i(t) = \frac{e^{\beta(\gamma_i - Q_i)}}{\sum_j e^{\beta(\gamma_j - Q_j)}}$$

β controls the exploration-exploitation trade-off. A small β leads to exploratory behavior, whereas a large β leads to a more exploitative behavior.

γ_i represents the alternative-specific bias. It captures preferences not explained by experience (like highways...)

5. **Why use RL models instead of RUM (Random Utility Models) like logit**

The choice rule looks almost identical. In RUM models we only get to see the final utility, while in RL we can see how the utility evolves each episode (hence how traveler learns).

RUM assumes an inertia. People tend to repeat past choices. But the author highlights that repetition may be a consequence of learning, rather than primitive behavioral trait.

6. Limitation of RL

RL only learns when there is feedback. If route A and route B both produce 20 min every single day then $Q_a = Q_b$ never changes. The RL predicts no learning, but real humans may still develop habits.